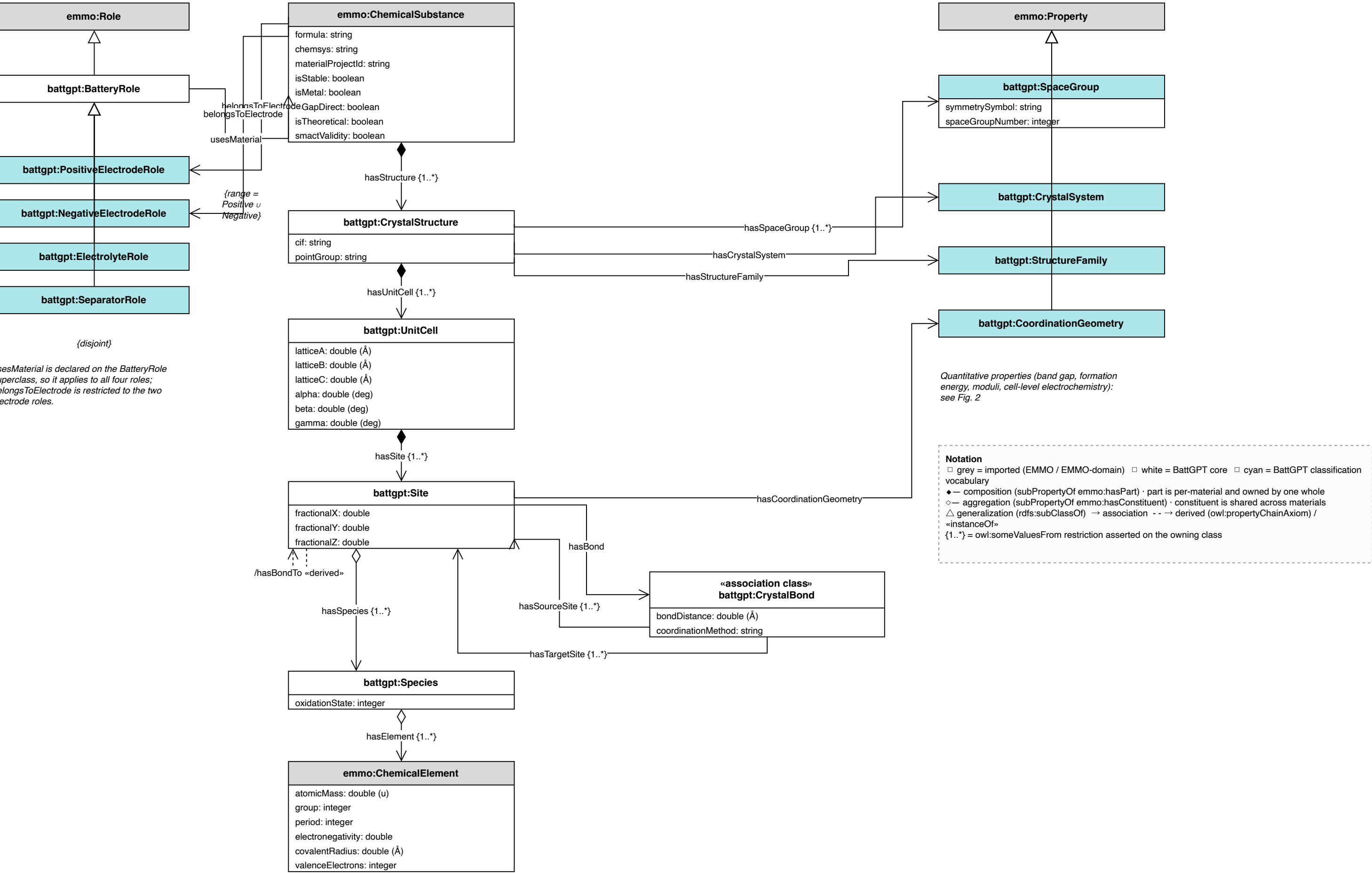
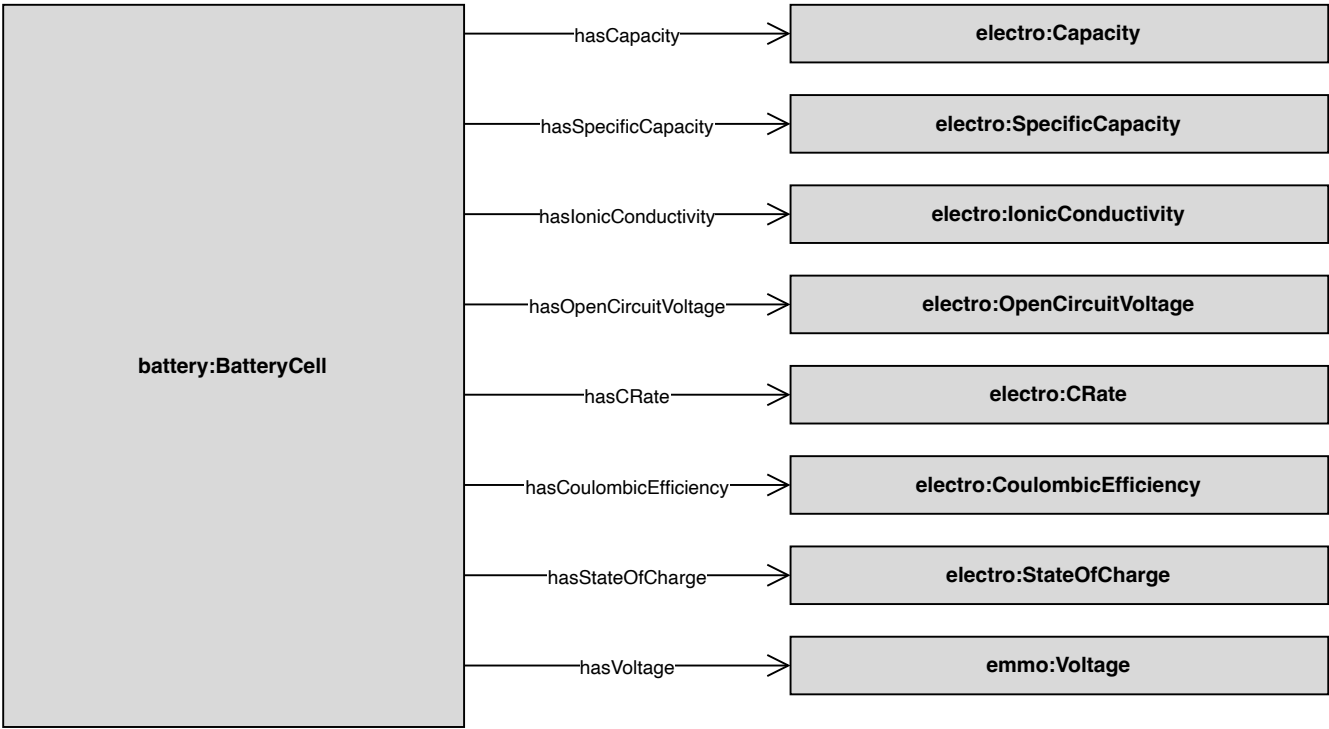
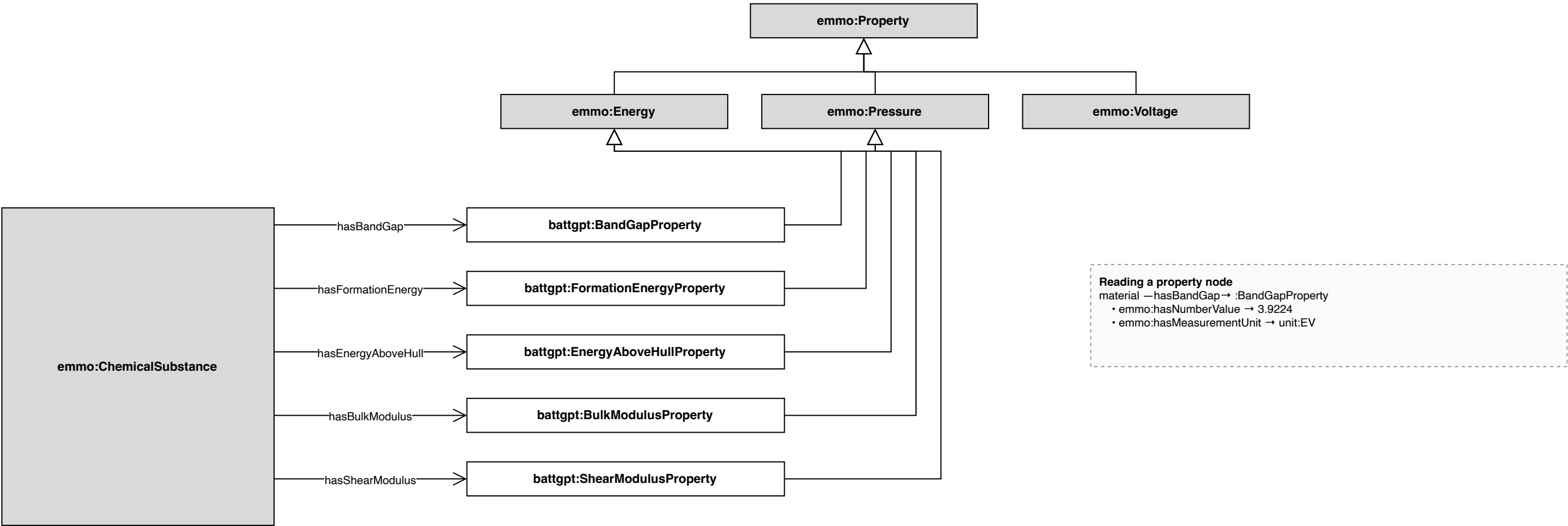
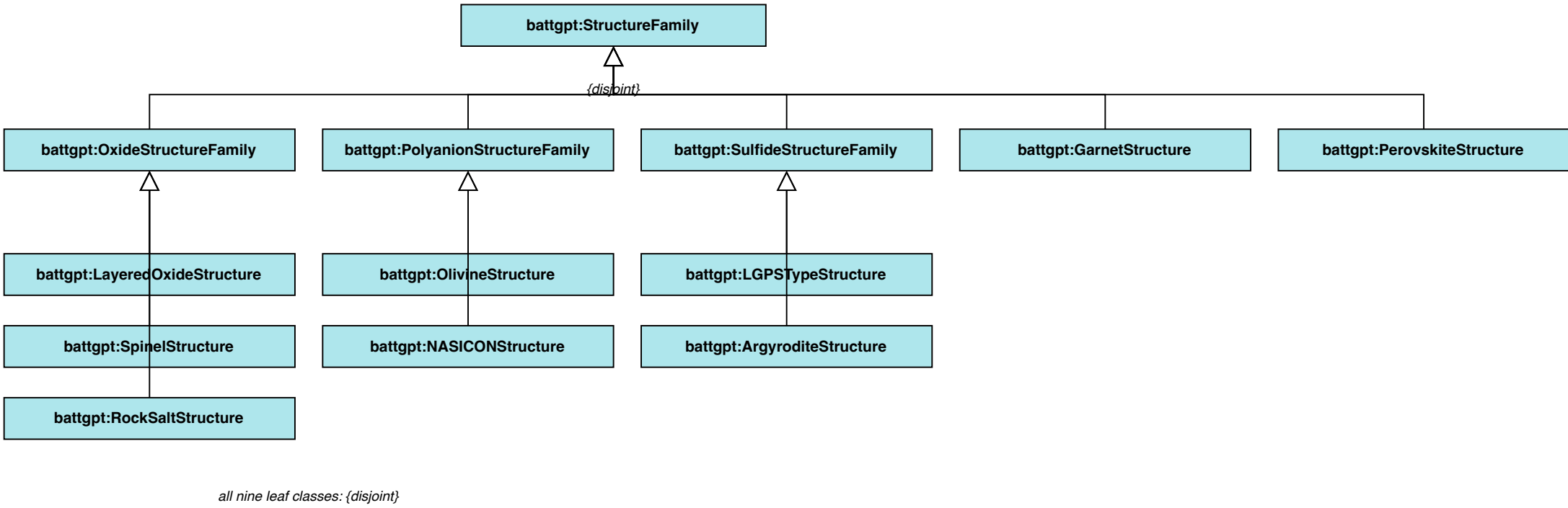


BattGPT Ontology — Core Class Model (M1)
material-centric: emmo:ChemicalSubstance is the hub; every structural class is reached from it





battery:BatteryCell and every electro: range class are reused from EMMO domain-battery 0.20.2 / domain-electrochemistry 0.37.2 — not redefined here. Only the eight has predicates are BattGPT's.*



Canonical individuals (owl:AllDifferent)

<u>LayeredOxideStructureIndividual</u> : StructureFamily	<u>SpinelStructureIndividual</u> : StructureFamily	<u>RockSaltStructureIndividual</u> : StructureFamily
<u>OlivineStructureIndividual</u> : StructureFamily	<u>NASICONStructureIndividual</u> : StructureFamily	<u>GarnetStructureIndividual</u> : StructureFamily
<u>PerovskiteStructureIndividual</u> : StructureFamily	<u>LGPSTypeStructureIndividual</u> : StructureFamily	<u>ArgyroditeStructureIndividual</u> : StructureFamily

battgpt:CoordinationGeometry

8 canonical individuals (owl:AllDifferent):
Linear · TrigonalPlanar · Tetrahedral ·
SquarePyramidal · Octahedral ·
PentagonalBipyramidal ·
SquareAntiprismatic · Cuboctahedral

battgpt:BatteryRole

4 canonical individuals (owl:AllDifferent):
PositiveElectrodeRoleIndividual ·
NegativeElectrodeRoleIndividual ·
ElectrolyteRoleIndividual · SeparatorRoleIndividual

BattGPT M0 Instance Model — LiFePO₄ (mp-19017)
values taken verbatim from the generated knowledge graph; dashed «instanceOf» links point to the M1 classes of Fig. 1

